Synthetic Phantom Validation Benchmark

Quantitative Optical-Flow & Strain Accuracy Evaluation against Mathematical Ground Truth

RESEARCH BENCHMARK NOTICE: This report evaluates algorithmic motion and strain estimation against an artificial synthetic phantom with known analytical ground truth. This synthetic experiment is strictly intended for algorithmic verification and software quality assurance. It does NOT establish clinical validity, diagnostic accuracy, or safety for human medical evaluation.

Phantom & Dataset Configuration

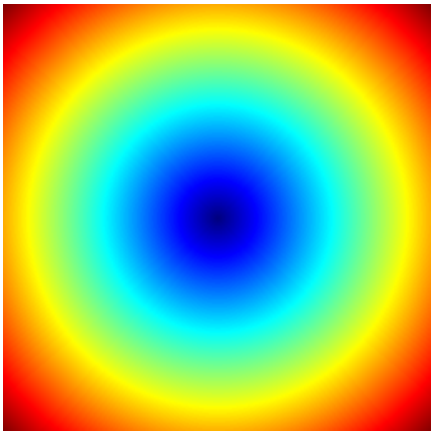
Dataset ID:	`94481cea-87e9-4d7e-8452-c7566eb03c7d`	**Deformation Mode:**	Radial Expansion
Dimensions:	256 × 256 px	**Deformation Strength:**	0.04
Myocardial Radii:	R_{in}=40.0 px, R_{out}=70.0 px	**Noise / Blur:**	Noise σ=0.0, Blur σ=0.0

Quantitative Accuracy vs Ground Truth

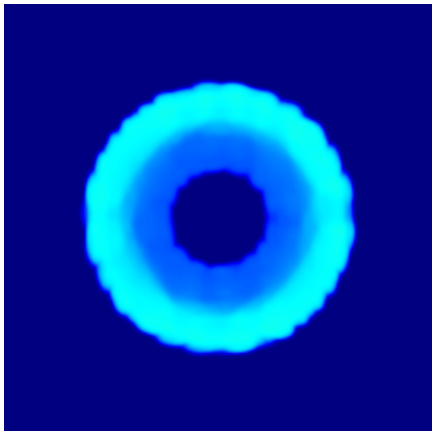
Measurement Category	Mean Abs Error (MAE)	RMSE	Pearson Correlation (r)	Endpoint Error (Mean / Max)
Displacement Field	0.1075 px	0.1284 px	0.9703	0.1370 px / 0.3294 px
Principal Strain 1	0.0227	0.0303	0.0000	Rel Err: 0.5674
Radial Deformation	0.0275	0.0321	0.0000	—
Circumferential Deform.	0.0132	0.0164	0.0000	—

Displacement Verification Maps

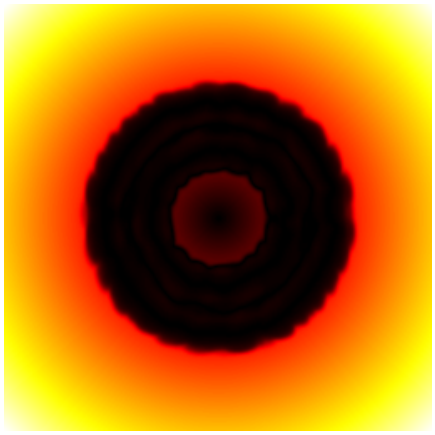
Ground Truth



Estimated



Absolute Error



Scientific Limitations:

- Synthetic deformation is mathematically controlled via affine and coordinate transformations.
- Synthetic texture is simplified and does not model speckle tracking ultrasound or MRI tagging physics.
- Optical flow measures apparent intensity motion, which may degrade under out-of-plane cardiac motion.
- Clinical imaging contains patient-specific anatomy, respiration, arrhythmias, and acoustic shadows.
- This benchmark serves to verify mathematical consistency and numerical stability of algorithms.